Notes: Bagwell and Staiger (AER, 1990)
A Theory of Managed Trade

Contents

1 One-sentence summary **3**

2 Big picture **3**
- 2.1 Core question . . . 3
- 2.2 Main answer . . . 3
- 2.3 What "managed trade" means here . . . 3
- 2.4 Key economic mechanism . . . 3
- 2.5 Main result . . . 3

3 Incremental contribution of the paper **3**
- 3.1 Explains special protection inside a cooperative-trade model . . . 3
- 3.2 Makes trade volume central . . . 4
- 3.3 Links managed trade to enforcement . . . 4
- 3.4 Distinguishes tariffs and quotas . . . 4
- 3.5 Reinterprets trade imbalances . . . 4
- 3.6 Foundation for later trade-agreement theory . . . 4

4 Conceptual framework **4**
- 4.1 Two countries and one sector . . . 4
- 4.2 Free trade volume . . . 4
- 4.3 Policy instruments . . . 4
- 4.4 Dynamic enforcement . . . 5
- 4.5 Why high volume is dangerous . . . 5

5 Model setup and derivations **5**
- 5.1 Trade condition under tariffs . . . 5
- 5.2 Static tariff best response . . . 5
- 5.3 Autarky equilibrium . . . 5
- 5.4 Defection from a cooperative tariff . . . 6
- 5.5 No-defection constraint . . . 6
- 5.6 Most cooperative tariff rule . . . 6
- 5.7 Quota rule . . . 6

6 Main cases in the paper **6**
- 6.1 Static tariff game . . . 6
- 6.2 Dynamic tariff game . . . 6
- 6.3 Dynamic quota game . . . 7

6.4 Trade balance extension . . . 7
6.5 Robustness and extensions . . . 7

7 Economic intuition **7**
7.1 Why special protection can be cooperative . . . 7
7.2 Why high trade volume raises defection incentives . . . 7
7.3 Why quotas differ from tariffs . . . 7
7.4 Why trade imbalance is not enough . . . 7

8 Identification and theoretical design **7**
8.1 What the paper identifies theoretically . . . 7
8.2 Key comparison . . . 8
8.3 Observable implications . . . 8
8.4 Important limitations . . . 8

9 Tables and figures **8**
9.1 Figure 1: Static tariff-game Nash equilibria; Figure 2: Static incentive to defect from a cooperative tariff . . . 8
9.2 Figure 3: Most cooperative tariff rule; Figure 4: Most cooperative quota rule . . . 9
9.3 Notes-created paired summary tables . . . 10

10 Short summary **10**

11 Conclusion **11**
11.1 Core conclusion . . . 11
11.2 Why the paper matters . . . 11
11.3 Policy implication . . . 11
11.4 Limitations . . . 11
11.5 Takeaway . . . 11

1 One-sentence summary

Bagwell and Staiger explain "managed trade" as a self-enforcing cooperative arrangement in which countries sustain low normal protection but raise special protection during trade-volume surges because high trade volume increases the short-run temptation to defect from liberal trade rules.

2 Big picture

2.1 Core question

Why do countries maintain relatively liberal trade policies in normal times while using "special" protection–such as tariffs, voluntary export restraints, and orderly market arrangements–when trade flows surge?

2.2 Main answer

The paper models trade policy as a repeated game. Cooperation is sustained by the threat of future punishment. When the underlying volume of free trade is high, the one-period gain from cheating on a cooperative tariff or quota is also high. To keep countries from defecting, the cooperative agreement itself must allow more protection in high-volume states.

2.3 What "managed trade" means here

Managed trade is not simply inefficient protection. It is a dynamic cooperative rule: low baseline protection in ordinary states plus state-contingent protection when trade volumes become unusually large. The purpose of special protection is to preserve the self-enforcing nature of the agreement.

2.4 Key economic mechanism

The incentive constraint has the form

$$\text{current gain from defection} \leq \text{discounted future loss from punishment.}$$

High trade volume raises the left-hand side. A higher cooperative tariff or tighter cooperative quota lowers the current gain from defection by reducing the volume of rents at stake.

2.5 Main result

A self-enforcing cooperative trade policy can be increasing in the natural volume of trade. Thus, trade liberalization and special protection can coexist in one theory: low protection is sustained when trade volume is normal; protection rises when trade volume is unusually high.

3 Incremental contribution of the paper

3.1 Explains special protection inside a cooperative-trade model

Before this paper, repeated-game models could explain why countries sustain lower tariffs than static Nash tariffs. Bagwell and Staiger add a theory of why countries sometimes raise protection while remaining in a cooperative regime.

3.2 Makes trade volume central

The paper's key state variable is not a political shock, a lobbying shock, or a terms-of-trade shock alone. It is the underlying free trade volume, V^f. The theory predicts that protection should respond to trade-flow surges because surges make defection more attractive.

3.3 Links managed trade to enforcement

The paper reframes special protection as an enforcement device. If the future punishment threat is not enough to sustain free trade in high-volume states, the cooperative agreement may need to restrict trade today to prevent a trade war tomorrow.

3.4 Distinguishes tariffs and quotas

The paper compares tariffs and quotas in a repeated setting. Quotas can create stronger defection incentives because a deviator can capture quota rents. This produces a different state-contingent cooperative rule, including discontinuities in the quota case.

3.5 Reinterprets trade imbalances

The model says that aggregate trade imbalances alone are insufficient to predict protection. The sectoral composition of trade-volume changes matters: protection rises when a country's imports or exports in a specific sector surge, not simply when the bilateral trade balance changes.

3.6 Foundation for later trade-agreement theory

The paper is an early step toward the Bagwell-Staiger view of trade agreements as solutions to international externalities and enforcement problems. It connects GATT-like cooperation, escape-clause logic, and self-enforcing agreements.

4 Conceptual framework

4.1 Two countries and one sector

The benchmark model has a home country and a foreign country. In each period, a shock determines the volume of trade that would occur under free trade. The domestic country is the exporter and the foreign country is the importer.

4.2 Free trade volume

The paper denotes the underlying free trade volume by

$$V^f = 1 - e,$$

where e is a random production shock. A lower e implies a larger natural export volume from the home country to the foreign country.

4.3 Policy instruments

Countries can use tariffs or quotas:

- In the tariff game, the exporter and importer choose taxes τ and τ^*.
- In the quota game, countries choose quantity restrictions.
- In both games, the key issue is whether the cooperative policy is self-enforcing.

4.4 Dynamic enforcement

Countries interact repeatedly. If no one defects, they follow the cooperative policy rule. If a country defects, both countries revert forever to a static Nash equilibrium. The threat of punishment sustains cooperation.

4.5 Why high volume is dangerous

When V^f is high, the amount of surplus and rent associated with trade is high. The deviating country can gain more by setting its best-response protection while the other country continues to cooperate. Hence high V^f tightens the incentive constraint.

5 Model setup and derivations

5.1 Trade condition under tariffs

Trade occurs if tariffs are not prohibitive:

$$\tau + \tau^* < \frac{2V^f}{\beta}.$$

This condition says that a larger free-trade volume can support larger tax wedges before trade is completely shut down.

5.2 Static tariff best response

When trade occurs, the domestic best response to the foreign tariff is

$$T_R(V^f, \tau^*) = \frac{2V^f}{3\beta} - \frac{\tau^*}{3}.$$

By symmetry, the interior Nash tariff is

$$\tau_N(V^f) = \frac{V^f}{2\beta}.$$

Detailed interpretation: The static Nash tariff rises with the underlying free trade volume. This is the static benchmark from which dynamic cooperation must depart.

5.3 Autarky equilibrium

The static tariff game also has a no-trade autarky equilibrium when tariffs are high enough. The quota game has autarky as the unique static Nash equilibrium.

Detailed interpretation: These static Nash outcomes serve as punishment equilibria in the repeated game. A harsher punishment makes cooperation easier to sustain.

5.4 Defection from a cooperative tariff

If the cooperative tariff is τ_C, a defecting country chooses

$$\tau_D(V^f, \tau_C) = \frac{2V^f}{3\beta} - \frac{\tau_C}{3}.$$

The static gain from defection is

$$\Omega(V^f, \tau_D(V^f, \tau_C), \tau_C) = W(V^f, \tau_D(V^f, \tau_C), \tau_C) - W(V^f, \tau_C, \tau_C).$$

5.5 No-defection constraint

A cooperative tariff function $\tau_C(V^f)$ must satisfy

$$\Omega(V^f, \tau_D(V^f, \tau_C), \tau_C) \leq \omega,$$

where ω is the discounted future loss from punishment.

Detailed interpretation: The right-hand side is determined by patience and the severity of future punishment. The left-hand side rises with free trade volume, so high-volume states require higher protection unless the punishment threat is very strong.

5.6 Most cooperative tariff rule

The paper characterizes the lowest nonnegative cooperative tariff that satisfies the no-defection constraint. The rule has three regions:

- free trade when V^f is low or moderate;
- positive tariffs when V^f becomes high enough;
- stronger protection for larger trade-volume surges.

5.7 Quota rule

In the quota game, the cooperative quota must also be self-enforcing. The key difference is that a deviator can capture quota rents. This makes the defection incentive larger, and the cooperative quota rule can display a discrete tightening when free trade volume crosses a threshold.

6 Main cases in the paper

6.1 Static tariff game

The static tariff game has two sets of Nash equilibria: an interior symmetric tariff equilibrium and a set of autarky equilibria. Figure 1 illustrates the best-response geometry.

6.2 Dynamic tariff game

With repeated interaction, countries can sustain more liberal policies than static Nash tariffs. But the cooperative tariff must be state-contingent: high V^f raises the temptation to defect, so the rule raises protection in high-volume states.

6.3 Dynamic quota game

The quota rule differs because quota rents create an additional object to capture through defection. As a result, the most cooperative quota may jump discontinuously at a threshold trade volume.

6.4 Trade balance extension

The model's trade-volume result does not mechanically translate into a simple trade-balance result. In multi-sector environments, trade balance may stay constant while sectoral trade volumes and protection change. The sectoral composition of trade flows is crucial.

6.5 Robustness and extensions

The paper discusses more general supply and demand functions, multiple sectors, trade imbalances, and asymmetric countries. The central presumption is preserved: when a country's incentive to defect rises with a trade-volume surge, self-enforcing cooperation requires more protection in that state.

7 Economic intuition

7.1 Why special protection can be cooperative

Special protection appears protectionist in a static model. In this paper, it can be part of cooperation: it lowers the one-period gain from cheating and thereby preserves future liberalization.

7.2 Why high trade volume raises defection incentives

When trade volume is high, tariff revenues and terms-of-trade effects are larger. A country can gain more by unilaterally raising protection while its partner remains cooperative. This makes the cooperative agreement fragile precisely when trade flows are largest.

7.3 Why quotas differ from tariffs

Tariff revenues under cooperation are shared symmetrically in the benchmark. Quota rents can be captured by a defector, which changes the incentive constraint. The same trade-volume shock therefore produces different optimal cooperative responses under tariffs and quotas.

7.4 Why trade imbalance is not enough

A bilateral trade deficit may widen because imports rise, or because exports fall. The model predicts higher protection when the relevant sectoral trade volume rises, not merely when the aggregate trade balance changes. Thus the sectoral composition of imbalance matters.

8 Identification and theoretical design

8.1 What the paper identifies theoretically

This is a theoretical paper, not an empirical causal design. The model isolates one mechanism: the interaction between trade-volume shocks and the incentive constraints of self-enforcing trade agreements.

8.2 Key comparison

The comparison is between:

- static Nash protection, where countries act noncooperatively each period;
- dynamic cooperation, where future punishment can discipline current choices;
- most cooperative state-contingent policy, where protection is only as high as needed to prevent defection.

8.3 Observable implications

The model predicts:

- low baseline protection in normal trade-volume states;
- temporary special protection when import or export volume surges;
- stronger special protection when surges are more unusual;
- different patterns for tariffs versus quotas;
- weak mapping from aggregate trade imbalances to protection unless sectoral composition is observed.

8.4 Important limitations

The model is stylized. It uses a partial-equilibrium setup, symmetric countries in the benchmark, complete information, and simple punishment rules. The paper does not estimate the magnitude of enforcement constraints, but it provides a theory of why observed protection is both low on average and episodically high.

9 Tables and figures

9.1 Figure 1: Static tariff-game Nash equilibria; Figure 2: Static incentive to defect from a cooperative tariff

Read: Figure 1 shows the static tariff best-response correspondences. The intersection I is the interior Nash tariff, while the upper-right region represents autarky equilibria. Figure 2 decomposes the static gain from unilateral tariff defection from a cooperative tariff.

Economic message: Figure 1 provides the punishment benchmark. Figure 2 explains why the incentive to defect rises with trade volume and falls when the cooperative tariff is higher.

Detailed interpretation: The two figures together set up the dynamic problem. Static Nash tariffs and autarky are inefficient relative to cooperation, but they are credible punishments. In the repeated game, countries choose the most liberal tariff that still makes defection unattractive.

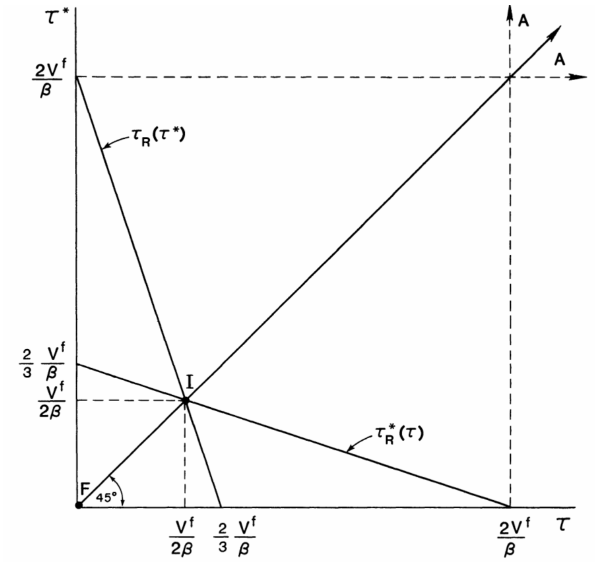

FIGURE 1

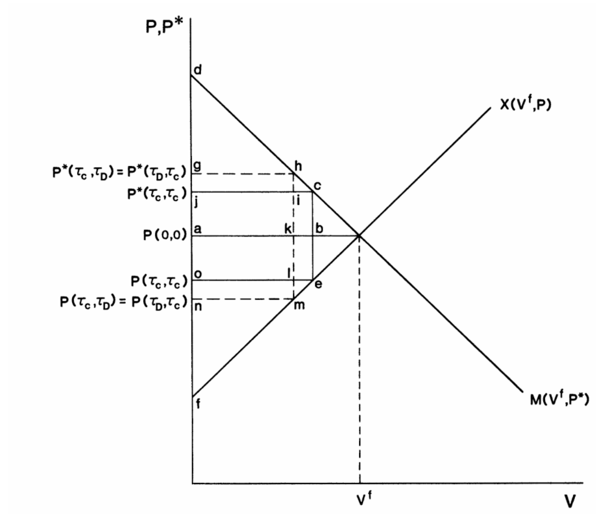

FIGURE 2

(a) **Figure 1: Static tariff-game Nash equilibria.** Interior Nash protection and autarky equilibria define the punishment alternatives in the repeated game.

(b) **Figure 2: Static incentive to defect.** The gain from defection is the extra tariff revenue net of the lost surplus from lower trade volume.

9.2 Figure 3: Most cooperative tariff rule; Figure 4: Most cooperative quota rule

Read: Figure 3 plots the most cooperative tariff and the resulting cooperative trade volume. The cooperative tariff is zero over a range of trade volumes, then rises when trade volume becomes high. Figure 4 plots the quota analogue. It includes a discontinuity: the cooperative quota tightens discretely when free-trade volume crosses the threshold.

Economic message: Figure 3 captures the managed-trade logic for tariffs: free trade in normal states, protection in high-volume states. Figure 4 shows that the same logic can be sharper under quotas because quota-rent capture changes the defection incentive.

Detailed interpretation: These are the central visuals of the paper. They translate an incentive constraint into a policy rule. Protection is not simply a static response to import competition; it is a dynamic response to the fragility of cooperation.

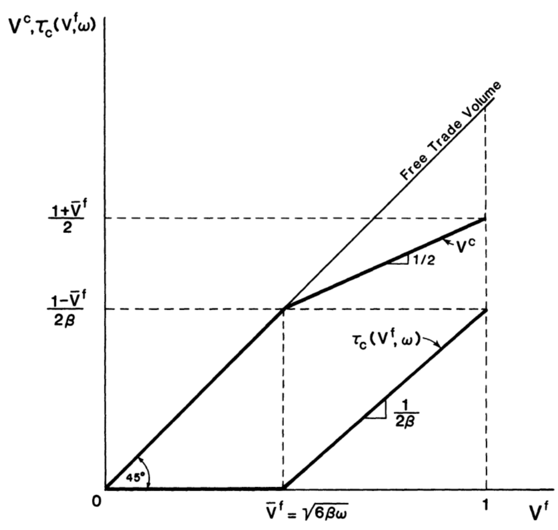

FIGURE 3

(a) **Figure 3: Most cooperative tariff rule.** Low or zero tariffs are feasible under normal trade volume; tariffs rise when trade volume is high enough to threaten cooperation.

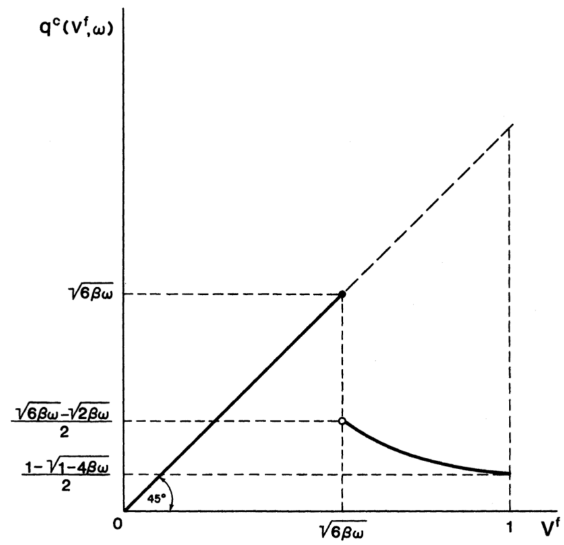

FIGURE 4

(b) **Figure 4: Most cooperative quota rule.** The quota rule can tighten discontinuously because defection can capture quota rents.

9.3 Notes-created paired summary tables

Object	**Role in model**
V^f	Underlying free trade volume; main state variable.
τ_N	Static Nash tariff, used as punishment benchmark.
$\tau_C(V^f)$	Cooperative tariff rule chosen to satisfy the incentive constraint.
τ_D	Best defection tariff from a cooperative policy.
Ω	Static gain from defection.

(a) **Model objects.** This notes-created table summarizes the notation used in the dynamic tariff game.

Prediction	**Economic interpretation**
Normal trade volume	Low or zero protection is self-enforcing.
Trade-volume surge	Higher protection is needed to deter defection.
More unusual surge	Stronger special protection.
Quota policy	Possible discrete tightening because rents matter.
Trade balance	Sectoral composition matters more than aggregate balance.

(b) **Main predictions.** This notes-created table summarizes the paper's comparative statics.

10 Short summary

- The paper explains managed trade with a repeated-game model of self-enforcing trade agreements.
- Static Nash protection is too high, but dynamic punishment can sustain more liberal policies.
- The temptation to defect rises when the underlying free trade volume is high.
- Therefore, cooperative protection can rise during trade-volume surges.
- Special protection can be a device for preserving cooperation rather than purely a breakdown of cooperation.
- Tariffs and quotas have different enforcement properties because quota rents change the incentive to defect.
- Aggregate trade balances alone do not determine protection; the sectoral structure of trade flows is crucial.

11 Conclusion

11.1 Core conclusion

Bagwell and Staiger show that managed trade can arise endogenously in a self-enforcing trade agreement. Countries cooperate by keeping protection low in normal periods, but they may need to raise protection when trade volumes surge because the temptation to defect becomes too strong.

11.2 Why the paper matters

The paper reconciles two postwar facts: declining baseline tariffs under GATT and persistent episodes of special protection. Rather than treating special protection as a simple failure of cooperation, the model shows how special protection may help sustain cooperation.

11.3 Policy implication

Escape clauses, voluntary export restraints, and other special instruments can be interpreted as enforcement devices. Their welfare implications depend on whether they prevent larger trade wars or merely transfer rents to protected sectors.

11.4 Limitations

The model is stylized and does not estimate enforcement constraints empirically. It abstracts from lobbying, distributional politics, imperfect monitoring, and asymmetric countries in the benchmark. Its value is the clean theoretical mechanism linking trade-volume volatility to state-contingent protection.

11.5 Takeaway

The central takeaway is that trade agreements must be robust to changing trade volumes. A rule that is liberal in normal times may not be self-enforcing during trade surges. Managed trade is the state-contingent response to that dynamic enforcement problem.